

Lab13 – CRD

As an application developer, you may want to install Kubernetes functionality that extends the platform using the Kubernetes operator pattern. The objective of this exercise is to familiarize yourself with creating and managing CRDs. You will not need to write a controller.

1. Create a CRD resource named `backup.example.com` with the following specification:
 - Group: `example.com`
 - Version: `v1`
 - Kind: `Backup`
 - Singular: `backup`
 - Plural: `backups`
 - Properties of type `string`: `cronExpression`, `podName`, `path`

```
vagrant@kubernetes-control-plane:~/lab$ vim backup-resource.yaml
vagrant@kubernetes-control-plane:~/lab$ cat backup-resource.yaml
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: backups.example.com
spec:
  group: example.com
  versions:
  - name: v1
    served: true
    storage: true
    schema:
      openAPIV3Schema:
        type: object
        properties:
          spec:
            type: object
            properties:
              cronExpression:
                type: string
              podName:
                type: string
              path:
                type: string
  scope: Namespaced
  names:
    kind: Backup
    singular: backup
    plural: backups
    shortNames:
    - bu
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f backup-resource.yaml
customresourcedefinition.apiextensions.k8s.io/backups.example.com created
vagrant@kubernetes-control-plane:~/lab$
```

```

vagrant@k8s-control-plane:~/lab$ kubectl api-resources | grep -i backup
backups          bu          example.com/v1      true      Backup
vagrant@k8s-control-plane:~/lab$

```

- Retrieve the details for the `Backup` custom resource created in the previous step.

```

vagrant@k8s-control-plane:~/lab$ kubectl get crd backups.example.com
NAME                CREATED AT
backups.example.com 2023-08-13T06:43:18Z
vagrant@k8s-control-plane:~/lab$ kubectl describe crd backups.example.com
Name:                backups.example.com
Namespace:
Labels:              <none>
Annotations:         <none>
API Version:         apiextensions.k8s.io/v1
Kind:                CustomResourceDefinition
Metadata:
  Creation Timestamp: 2023-08-13T06:43:18Z
  Generation:        1
  Resource Version:   57838
  UID:               8048d87f-e5ed-471e-9acc-514d0029809a
Spec:
  Conversion:
    Strategy: None
  Group:      example.com
  Names:
    Kind:      Backup
    List Kind: BackupList
    Plural:    backups
    Short Names:
      bu
    Singular:  backup
  Scope:      Namespaced
  Versions:
    Name: v1
    Schema:
      openAPIV3Schema:
        Properties:
          Spec:
            Properties:

```

- Create a custom object named `nginx-backup` that uses the CRD kind `Backup`.

Provide the following property values:

- `cronExpression`: `0 0 \* \* \*`
- `podName`: `nginx`
- `path`: `/usr/local/nginx`

```

vagrant@k8s-control-plane:~/lab$ vim backup.yaml
vagrant@k8s-control-plane:~/lab$ cat backup.yaml
apiVersion: example.com/v1
kind: Backup
metadata:
  name: nginx-backup
spec:
  cronExpression: "0 0 * * *"
  podName: nginx
  path: /usr/local/nginx
vagrant@k8s-control-plane:~/lab$ kubectl apply -f backup.yaml
backup.example.com/nginx-backup created
vagrant@k8s-control-plane:~/lab$
vagrant@k8s-control-plane:~/lab$

```

4. Retrieve the details for the `nginx-backup` object created in the previous step.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl get backups
NAME          AGE
nginx-backup  3m3s
vagrant@kubernetes-control-plane:~/lab$ kubectl describe backup nginx-backup
Name:          nginx-backup
Namespace:     default
Labels:        <none>
Annotations:    <none>
API Version:   example.com/v1
Kind:          Backup
Metadata:
  Creation Timestamp:  2023-08-13T06:49:28Z
  Generation:         1
  Resource Version:    58407
  UID:                 cd5b7fd6-ad67-4c78-99b3-154e8ada3c47
Spec:
  Cron Expression:     0 0 * * *
  Path:                /usr/local/nginx
  Pod Name:            nginx
Events:               <none>
vagrant@kubernetes-control-plane:~/lab$
vagrant@kubernetes-control-plane:~/lab$
```